

Report on the Threat of Recent DDoS Attacks

Introduction

Distributed Denial of Service (DDoS) attacks continue to evolve, leveraging vast botnets and advanced multi-vector strategies to disrupt major online infrastructures. In recent years (2024–2025), DDoS threats have grown both in scale and sophistication, with incidents targeting telecommunication providers, cloud services, financial institutions, and public infrastructure. This report analyzes five major DDoS attacks and selects one for deeper investigation.

Overview of Recent DDoS Attacks

1. Cloudflare 11.5 Tbps Attack (September 2025)

- **Target:** Cloudflare's global content delivery and protection infrastructure.
- **Technology Used:** HTTP/2 "Rapid Reset" multi-vector flooding method generating 11.5 terabits per second.
- **Motive:** Likely botnet test and extortion attempt.
- **Impact:** Temporary strain on Cloudflare's routing nodes; no major downtime reported.
- **Mitigation:** Adaptive traffic filtering, load balancing across global data centers, and rapid HTTP/2 rate limiting.

2. Aisuru Botnet Attack on U.S. ISPs (May 2025)

- **Target:** Multiple U.S. internet service providers including KrebsOnSecurity.
- **Technology Used:** Aisuru botnet leveraging residential proxies, emitting over 6.35 Tbps of attack traffic.
- **Motive:** Demonstration of "botnet-for-hire" services and criminal advertisement.

- **Impact:** Packet loss and routing slowdowns affecting hundreds of thousands of users in the U.S.
- **Mitigation:** ISP-level peering filters, subnet isolation, and anomaly detection systems.

3. India's Largest DDoS Attack (July 2025)

- **Target:** Indian government and telecom service domains.
- **Technology Used:** Multi-vector network flood exploiting IoT devices and reflection vectors (UDP/DNS amplification).
- **Motive:** Political disruption and digital protest activity.
- **Impact:** Nationwide service instability with millions of connection requests dropped.
- **Mitigation:** Strengthening edge firewalls, DNS security extensions, and centralized monitoring networks.

4. European Financial Service Attack (2024)

- **Target:** European financial exchange platform handled by Akamai Prolexic.
- **Technology Used:** Layer 3/4 UDP flood combined with SYN+PSH amplification lasting nearly a full day.
- **Motive:** Hacktivist retaliation over economic sanctions; potentially also ransom-driven.
- **Impact:** Trading interruptions and data synchronization delays.
- **Mitigation:** Upstream traffic scrubbing and dynamic capacity scaling.

5. Anonymous Sudan Attack on Microsoft (June 2023)

- **Target:** Microsoft 365 and Azure service portals.
- **Technology Used:** Layer 7 HTTP flood via slow request handling and exploitation of caching layers.
- **Motive:** Political hacktivism related to U.S. foreign policy.
- **Impact:** Temporary global service denial affecting users across OneDrive, Outlook, and Teams.
- **Mitigation:** Traffic filtration through CDNs, API throttling, and improved global redundancy.

Selected Incident for Investigation: Cloudflare 11.5 Tbps DDoS Attack (2025)

Target

The attack was aimed at Cloudflare, a major internet security and performance provider that protects a significant portion of global online services. The objective was to overwhelm Cloudflare's global edge infrastructure to cause large-scale unavailability.

Technology Used

This DDoS assault utilized a method based on the **HTTP/2 Rapid Reset vulnerability**, where a massive number of simultaneous requests were rapidly canceled and retried. The traffic peaked at **11.5 terabits per second**, making it the most extensive DDoS bandwidth recorded in history.

Attacker's Motive

Analysts attribute the motive to either **a demonstration of capability by a new DDoS-as-a-service botnet** or a **financial extortion attempt**. The event also appeared to serve as a benchmark for further DDoS-for-hire advertisement campaigns.

Overall Impact

Due to Cloudflare's strong autonomous mitigation capabilities, **no long-term disruption** occurred, though global backbone routers experienced temporary latency spikes. The incident emphasized the growing ability of attackers to weaponize modern web protocols.

Defensive Strategies

- Enforce rate-limiting rules against HTTP/2 rapid resets.
- Deploy distributed load-balancing and multi-region CDN scrubbing centers.
- Monitor for abnormal connection resets and spikes in simultaneous requests.
- Maintain layered defenses supported by AI-based behavior analysis.

Conclusion

DDoS attack trends in 2024–2025 reveal increasing dependence on high-bandwidth and protocol-specific exploitation. Attack motives now span from political activism to financial extortion, making the threat both versatile and unpredictable. Effective defense requires **global collaboration, autonomous response mechanisms, and continuous AI-driven anomaly detection** to preemptively neutralize threats before service disruption occurs.
